CLINICAL · NOSE & SINUSES

# Choanal atresia and congenital nasal masses

Newborns breathe through the nose. A baby who turns blue when quiet or feeding and pink when crying has bilateral choanal atresia until proven otherwise, and needs an oral airway now. A midline mass on a child's nose may connect to the brain: it is imaged, never biopsied or probed, before a plan is made.

**AT A GLANCE**

- Bilateral choanal atresia is an airway emergency: oral airway, then repair.
- Unilateral atresia presents later with one-sided discharge.
- Dermoid, glioma and encephalocele: image before biopsy.
- Furstenberg sign suggests encephalocele.
- Stents after choanal atresia repair do not reduce revision; piriform aperture stenosis needs pituitary assessment.

## 1 CHAPTER ONE Overview

Developmental anomalies of the nose.

### 1.1 The conditions

**CONGENITAL NASAL ANOMALIES**

- AIRWAY **Choanal atresia** pyriform aperture stenosis, atresia of the nares
- MIDLINE MASSES **Nasal dermoid** glioma, encephalocele
- CYSTS **Nasolacrimal duct cyst** nasoalveolar cyst
- OTHER **Haemangioma** proboscis lateralis, arhinia (rare)

### 1.2 Why it matters

Obligate nasal breathing in the first weeks of life makes bilateral obstruction an emergency; midline masses carry a risk of meningitis and CSF leak.

# 2 CHAPTER TWO Choanal atresia

The closed back door.

## 2.1 Embryology

Persistence of the *bucconasal membrane* (or abnormal mesoderm) between the nasal cavity and the nasopharynx.

## 2.2 Types

- One-sided or both; one-sided is commoner and may go unnoticed until adulthood.
- Complete or incomplete.
- Bony (about 90% in Dhingra; modern series show most are mixed bony and membranous) or purely membranous.

## 2.3 Associations

Associated anomalies are common, especially with bilateral atresia: *CHARGE* syndrome (coloboma, heart defects, choanal atresia, retarded growth and development, genital and ear anomalies), Treacher Collins and Crouzon syndromes.

## 2.4 Clinical picture

### 2.4.1 Bilateral

Respiratory distress at birth: *cyclical cyanosis*, blue at rest or when feeding and pink when crying (the baby breathes through the mouth while crying), with difficulty feeding.

### 2.4.2 Unilateral

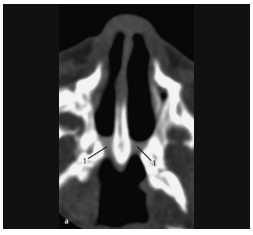
One-sided nasal obstruction and persistent thick discharge from that side, often discovered in later childhood.

## 2.5 Diagnosis

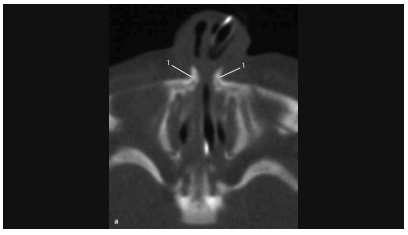
- Mucoid discharge without air bubbles.
- Failure to pass a soft catheter (a size 6 to 8 French) more than about 3 cm into the nose.
- Methylene blue in the nose fails to reach the pharynx; a wisp of cotton under the nostril does not move.
- Flexible nasal endoscopy.
- **CT** in the axial plane is the definitive test: it shows the atretic plate, narrowing of the posterior nasal cavity and thickening of the vomer, and separates bony from membranous atresia.

**IMAGING IN THE APP** 2 cases

Choanal atresia and pyriform aperture stenosis on CT.



Choanal Atresia
Congenital · 10 images



Piriform Aperture Malformation
Congenital · 2 images

## 2.6 Management

### 2.6.1 Emergency airway in bilateral atresia

An oral airway, or a feeding nipple with the end cut off taped in the mouth (*McGovern's technique*), keeps the mouth open and avoids tracheostomy; orogastric feeding. Intubation if needed.

**NEWBORN WITH CYCLICAL CYANOSIS**

1. Insert an oral airway or McGovern nipple.
2. Try to pass a catheter through each side of the nose.
3. Arrange CT of the nasal cavity after suction and decongestion.

4 Look for CHARGE anomalies: heart, eyes, ears, genitalia.

#### 2.6.2 Definitive repair

- **Endoscopic transnasal repair** with a drill: removal of the atretic plate and the posterior part of the vomer, with or without mucosal flaps and stents. The standard approach now.
- **Transpalatal repair:** better exposure, historically at about 18 months, but risks palatal growth disturbance.
- Unilateral atresia can be repaired electively in childhood.

#### 2.6.3 After repair

Restenosis is common, so regular review, dilatation and revision may be needed.

**OPERATIVE GUIDES IN THEATRE** 2 operations

**Choanal Atresia/Pyriform Aperture Stenosis**
Choanal atresia and pyriform aperture stenosis

**Midline Nasal Masses**
Midline nasal masses

## 2.7 Pyriform aperture stenosis

Narrowing of the bony nasal entrance from overgrowth of the nasal processes of the maxilla, presenting like bilateral choanal atresia but at the front of the nose. A single central maxillary incisor suggests holoprosencephaly and pituitary problems. Mild cases are managed medically; severe cases by drilling through a sublabial approach.

# 3 CHAPTER THREE Congenital midline nasal masses

Dermoid, glioma and encephalocele.

## 3.1 Embryology

In early development a projection of dura passes through the *foramen caecum* (between the frontal and ethmoid bones) into the prenasal space, or between the frontal and nasal bones through the *fonticulus frontalis*. Normally it regresses. If it

does not, ectodermal elements (a dermoid), isolated glial tissue (a glioma) or a herniation of brain and meninges (an encephalocele) remain, sometimes with an intracranial connection. Incidence is about 1 in 20,000 to 40,000 births.

## 3.2 Nasal dermoid

### 3.2.1 Picture

The commonest midline nasal mass: a firm, non-tender, non-pulsatile, non-compressible swelling anywhere from the glabella to the columella, often with a *midline pit* that may sprout a hair. It may become infected repeatedly.

### 3.2.2 Types

- Simple dermoid beneath the skin, in front of the nasal bones, without an opening.
- Dermoid with a sinus, whose tract runs under the nasal bones to the upper septum.
- Dermoid with a tract through the foramen caecum or cribriform plate to the dura, or intradural extension, which risks meningitis.

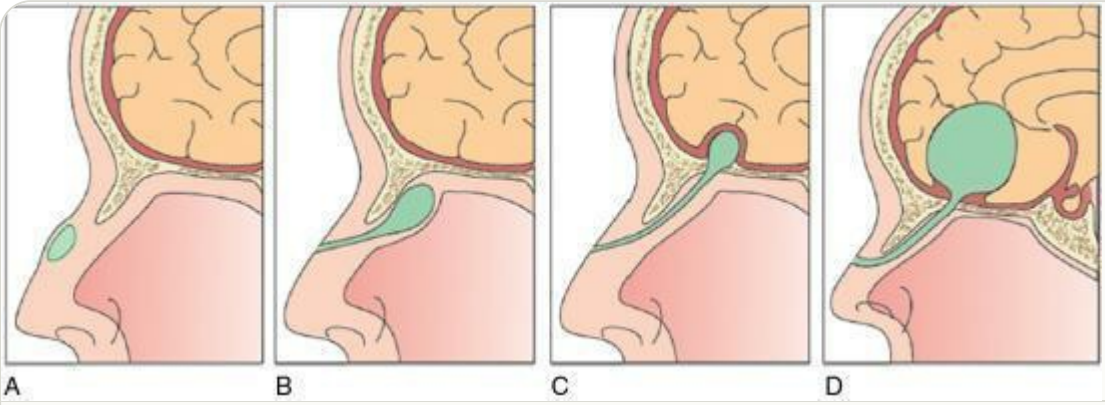



**FIGURE 1** Types of nasal dermoid: (A) simple, (B) with an external pit, lying deep to the nasal bones, (C) with a connection to the dura, (D) intradural.
Dhingra, fig. 25.3

### 3.2.3 Treatment

Complete excision of the cyst, the tract and any involved bone, cartilage and septum, often by an open rhinoplasty approach; combined neurosurgical repair when the tract reaches the dura.

## 3.3 Nasal glioma

### 3.3.1 Picture

A mass of ectopic glial tissue, not a true tumour, sometimes called a nasal glial heterotopia. About 60% are extranasal (firm, reddish swelling on the bridge or side of the nose with telangiectasias), 30% intranasal (a pale, firm polypoid mass, mistaken for a polyp) and 10% both. About 15 to 20% have a fibrous stalk to the dura. Non-pulsatile and non-compressible.



**FIGURE 2** Extranasal glioma on the right side of a child's nose.
Dhingra, fig. 25.4

### 3.3.2 Treatment

Complete excision by an external approach; craniotomy if there is an intracranial connection.

## 3.4 Encephalocele

### 3.4.1 Picture

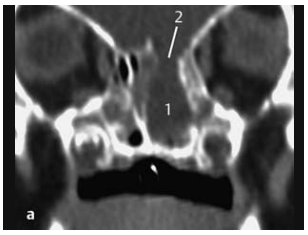
A herniation of meninges (meningocele), or meninges with brain (meningoencephalocele), through a skull base defect, communicating with the subarachnoid space. It is soft, bluish, *pulsatile* and *compressible*, enlarges on crying, straining or jugular compression (*Furstenberg sign*), and may transilluminate.

### 3.4.2 Types by site

- **Sincipital (frontoethmoid), presenting outside the nose:** nasofrontal (glabella), nasoethmoidal (side of the nose) and naso-orbital (medial orbit).
- **Basal, presenting inside the nose or nasopharynx:** transethmoidal, sphenoethmoidal, transsphenoidal and sphenomaxillary, often mistaken for a polyp.

**IMAGING IN THE APP** 1 case

Meningoencephalocele through the skull base.




Meningoencephalocele

Congenital · 7 images

### 3.4.3 Treatment

Neurosurgical or endoscopic: dividing the stalk, excising the extracranial part and repairing the dural and bony defect.

**IMAGE BEFORE YOU TOUCH**

A midline nasal mass in a child may connect with the brain. Biopsy or simple removal can cause CSF leak and meningitis. Obtain MRI (with CT for the bone) first.

**TELLING THE MIDLINE MASSES APART**

| | DERMOID | GLIOMA | ENCEPHALOCELE |
|---|---|---|---|
| **Consistency** | Firm | Firm | Soft |
| **Pulsatile or compressible** | No | No | Yes |
| **Furstenberg sign** | Negative | Negative | Positive |
| **Pit or hair** | Often | No | No |
| **Intracranial connection** | Sometimes | About 15 to 20%, fibrous stalk | Always, with CSF |

# 4 CHAPTER FOUR Other congenital anomalies

Less common.

## 4.1 Nasolacrimal duct cyst

A cystic dilatation of the lower nasolacrimal duct when Hasner's valve fails to open, seen as a bluish swelling under the inferior turbinate; bilateral cysts can obstruct a newborn's breathing. Endoscopic marsupialisation.

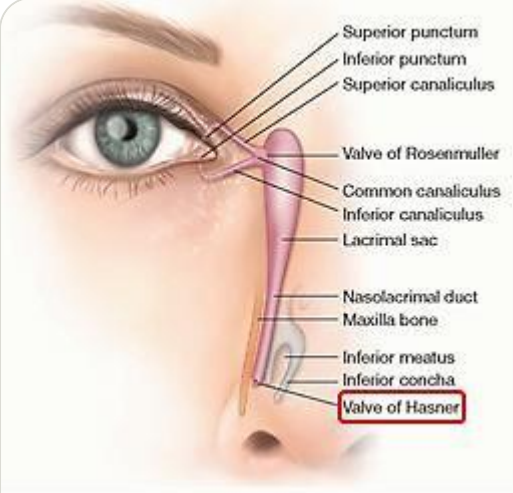



**FIGURE 3** The lacrimal drainage system and Hasner's valve in the inferior meatus.
Anatomy atlas · ENT Steps p. 100

## 4.2 Haemangioma

A strawberry swelling of the nasal tip that grows in the first year and regresses over several years; propranolol for problem lesions.

## 4.3 Rare anomalies

Arhinia (absent nose), proboscis lateralis, bifid nose and congenital atresia of the nares.

**CURRENT PAEDIATRIC PRACTICE (OXFORD TEXTBOOK)**

Neonates breathe preferentially, not obligatorily, through the nose for the first weeks to months, which is why bilateral nasal obstruction presents with distress at birth. About half of children with CHARGE syndrome have bilateral choanal atresia, and craniosynostoses such as Crouzon and Apert are rarer associations. Unilateral atresia is commoner, usually isolated and presents later with a one-sided discharge; CT is best kept for planning once surgery is decided, to avoid repeated radiation and sedation.

**Stents.** After endoscopic repair of bilateral atresia some surgeons stent for two to six weeks with cut endotracheal tubes, but recent series show no difference in revision rates with or without stents, and because stents need intensive care by parents and sometimes removal under anaesthesia, their use is declining; unilateral stenting is not worthwhile. Postoperative suction, saline and topical steroids reduce granulation and early scarring.

**Piriform aperture stenosis**, an overgrowth of the nasal process of the maxilla, mimics bilateral choanal atresia (a catheter cannot be passed) but is anterior; it may accompany a single central maxillary incisor and holoprosencephaly, so pituitary and brain assessment is needed. Endoscopy and CT distinguish the two. Surgery, a sublabial drilling of the aperture, is reserved for persistent respiratory distress or problems with sleep, growth or feeding. **Mid-nasal stenosis** can coexist. **Septal surgery** in children was long avoided for fear of affecting facial growth, but conservative septoplasty for significant obstruction appears safe. Adenoids are largest in absolute terms at 6 to 10 years, but largest relative to the nasopharynx, and so most obstructive, at 2 to 6 years.

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Choanal atresia:** embryology, cyclical cyanosis, tests, McGovern nipple, repair.
- **CHARGE syndrome.**
- **Dermoid, glioma and encephalocele:** embryology and differences.
- **Furstenberg sign.**

- **Why not to biopsy** a midline nasal mass.

## 5.2 Common pitfalls

COMMON PITFALLS

- Missing unilateral atresia in a child with one-sided discharge.
- Taking a nasal mass in a child for a polyp.
- Forgetting the associated anomalies of CHARGE.

Written for this app from Oxford Textbook of Otolaryngology ch. 78, Oxford Specialist Handbook ch. 23, Oxford Handbook of ENT ch. 17, Dhingra ch. 25, 29 and K. J. Lee ch. 31, 53. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

# On imaging

*The imaging of this topic: key findings, differentials and representative scans.*

NOSE & NASAL CAVITY · CONGENITAL

## Choanal Atresia

### IMAGING FINDINGS

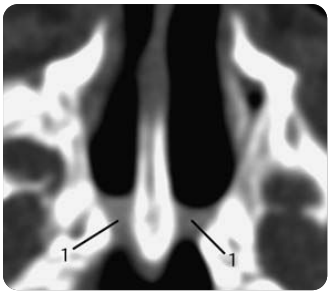
- Narrowed posterior choanae
- Thickened vomer; medially bowed posterior maxilla
- Bony, membranous or mixed closure; fluid in the nose

### TOP DIFFERENTIALS

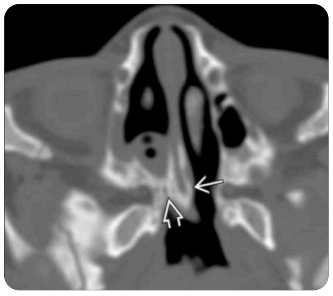
- Choanal stenosis
- Pyriform aperture stenosis
- Nasolacrimal duct mucocele

### CHECKLIST

- CT after suction and decongestion
- CHARGE: look at the inner ears




CT, axial. Both choanal areas demonstrate narrowing of the bony passage, but congenital fibrous closure (1) is the cause of the distres…



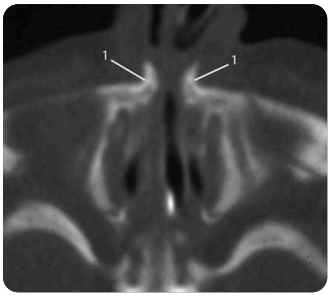
Axial bone CT through the upper choana in a child with mixed bony/osseous choanal atresia shows osseous choanal obstruction…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 670.

NOSE & NASAL CAVITY · CONGENITAL

## Piriform Aperture Malformation

- A developmental disorder of this kind may result in concomitant pathologies, such as the fusion of both maxillary central incisors to form one megaincisor (Fig. 7.15), in the majority of patients. Holoprosencephaly and abnormalities of the pituitary–adrenal axis are also encountered.




CT, axial. The severe dyspnea was due to gross narrowing of the anterior nasal passage caused by bilateral bony stenosis of the piriform aperture (1)….

NOSE & NASAL CAVITY · CONGENITAL

# Meningoencephalocele

### IMAGING FINDINGS

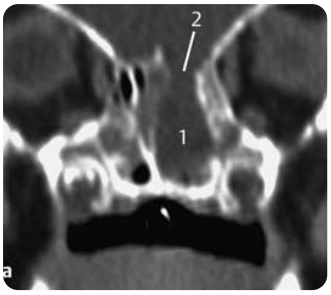
- Bony defect in the cribriform plate, lateral lamella, ethmoid roof or sphenoid
- Soft tissue or fluid through the defect
- MRI: CSF, meninges or brain in continuity with the intracranial contents

### TOP DIFFERENTIALS

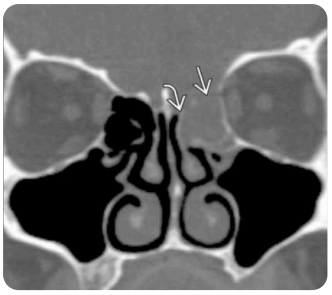
- Skull base defect without leak
- Sinus opacification from mucus

### CHECKLIST

- Lateral sphenoid recess and multiple defects
- Signs of raised intracranial pressure: empty sella




CT, coronal. The left nasal cavity is filled with a soft-tissue mass (1) that seems to be connected to brain through a defect in the roof of the…

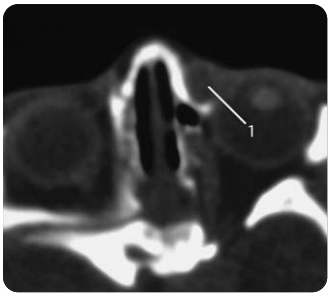


Coronal bone CT shows large bony defect in left ethmoid roof, lateral to insertion of middle turbinate. Because there is complete opacification of the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 920.

ORBIT & SINUS RELATIONS · CONGENITAL

# Nasolacrimal Mucocele




Patient with frequent and fluctuating subcutaneous swellings below the left medial canthus.
Dunnebier, Imaging for…